

EXP.NO-6	Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.

AIM:

Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.

ALGORITHM:

Input:

The user enters username and password in the login form and submits it.

1.Create the Login Page:

Design an HTML page with input fields for username and password.

Add a submit button.

Set the form action to the Login Servlet and method to POST.

2.Create the Login Servlet:

Create a Servlet class that extends HttpServlet.

Override the doPost() method.

Retrieve username and password using request.getParameter().

3.Validate User:

Check the entered credentials with predefined values.

If valid, create a session using HttpSession.

Store user data using session.setAttribute().

4.Redirect to Home Page:

After successful login, redirect the user to the Home Servlet.

5. Session Tracking:

In Home Servlet, retrieve session using request.getSession(false).

Check if session exists and user attribute is present.

6.Display Output:

If session exists, display welcome message with username.

If not, display "Please login first".

7. Run the Application:

Deploy the project on Apache Tomcat server.

Open login page in browser and test login functionality.

CODE:

login.html

```
<!DOCTYPE html>

<html>

<head>

    <title>TechScope Login</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="card">

    <h1 class="title">TechScope</h1>

    <p style="color:#64748b; font-size:14px;">Login to continue</p>

    <form action="login" method="post">

        <input type="text" name="username" placeholder="Username">

        <input type="password" name="password" placeholder="Password">

        <button type="submit">Login</button>

    </form>

</div>

</body>
```

</html>

loginservlet.java

```
package com.techscope;
```

```
import java.io.*;
```

```
import jakarta.servlet.*;
```

```
import jakarta.servlet.http.*;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
@WebServlet("/login")
```

```
public class LoginServlet extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        String user = request.getParameter("username");
```

```
        String pass = request.getParameter("password");
```

```
        response.setContentType("text/html");
```

```
        PrintWriter out = response.getWriter();
```

```
        String[][] users = {
```

```
            {"admin", "1234"},
```

```
            {"user136", "pass123"},
```

```
            {"user1", "1111"}
```

```
        };
```

```
        boolean valid = false;
```

```
        for(String[] u : users) {
```

```
            if(u[0].equals(user) && u[1].equals(pass)) {
```

```
                valid = true;
```

```

        break;
    }
}

if(valid) {

    HttpSession session = request.getSession();

    session.setAttribute("user", user);

    response.sendRedirect("home");

} else {

    out.println("<h2 style='color:red;'>Invalid Login</h2>");

}

}

}

```

HomeServlet.java

```

package com.techscope;

import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.WebServlet;

@WebServlet("/home")

public class HomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        response.setContentType("text/html");
    }
}

```

```
PrintWriter out = response.getWriter();

out.println("<html>");
out.println("<head>");
out.println("<link rel='stylesheet' href='style.css'>");
out.println("</head>");
out.println("<body>");

if(session != null && session.getAttribute("user") != null) {

    String user = (String) session.getAttribute("user");
    out.println("<div class='card'>");
    out.println("<h1 class='title'>TechScope</h1>");
    out.println("<h2>Welcome, " + user + "</h2>");
    out.println("<p>You are now logged in.</p>");
    out.println("</div>");

} else {

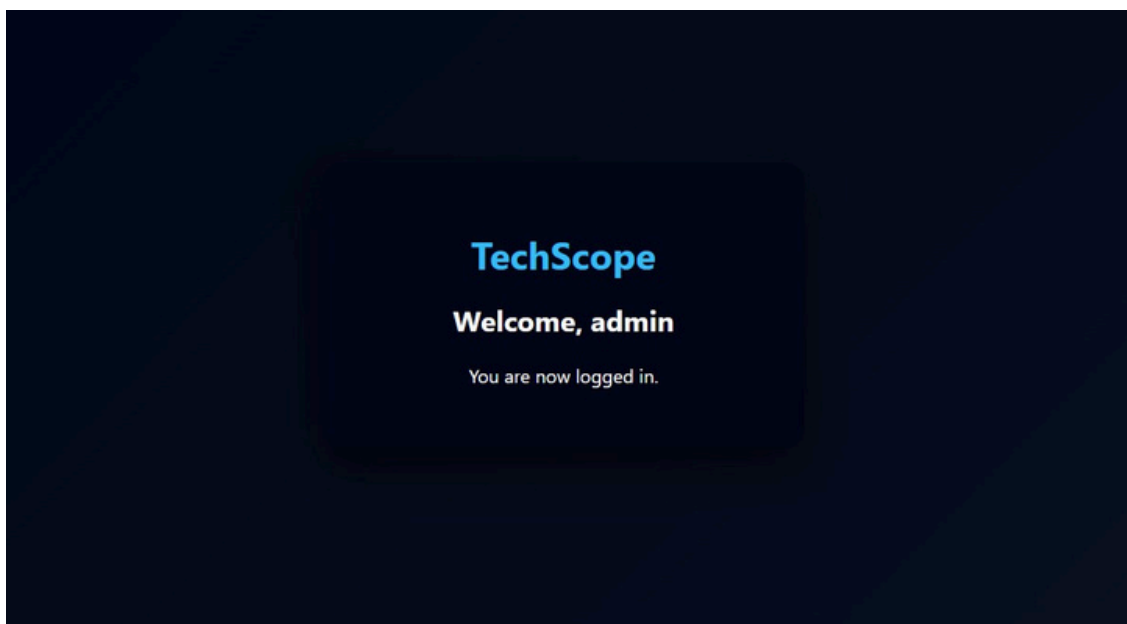
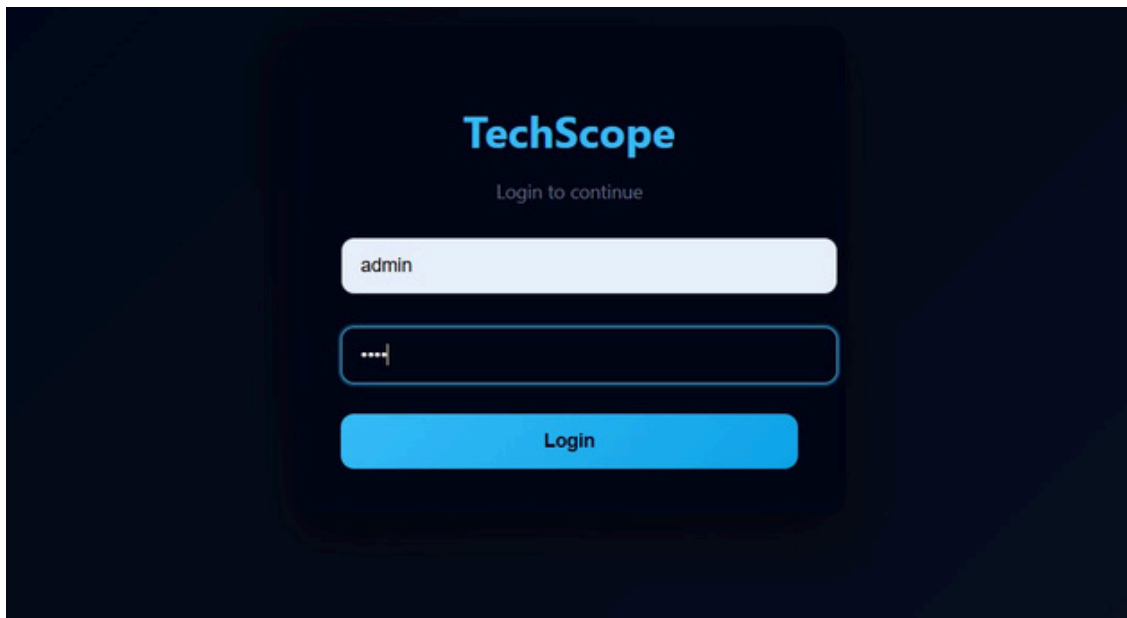
    out.println("<div class='card'>");
    out.println("<h2>Please login first!</h2>");
    out.println("<a href='login.html'>Go to Login</a>");
    out.println("</div>");

}

out.println("</body>");
out.println("</html>");

}
```

OUTPUT:



RESULT:

Thus, session tracking was successfully implemented using HttpSession. The user's login state was maintained across multiple requests, and access to the home page was controlled based on session availability.